

BREAK EVEN POINT ANALYSIS (30060 – MANAGEMENT)

CONTENTS & READINGS

Walsh Book

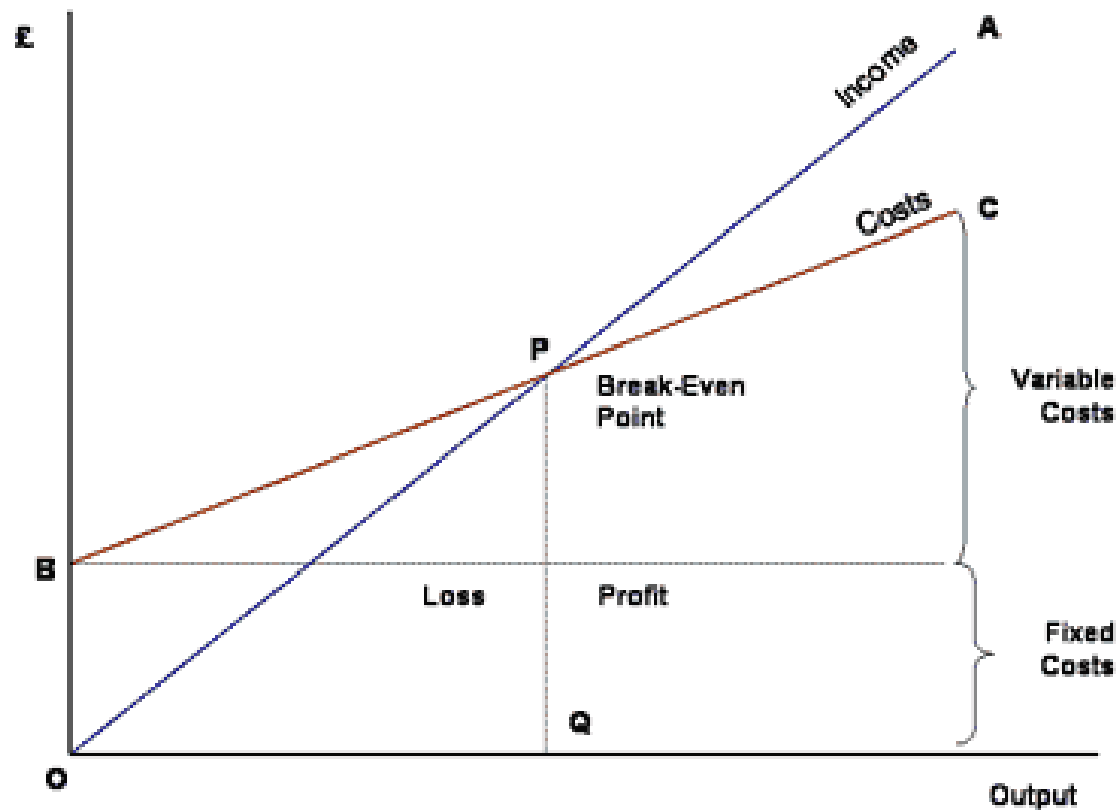


AGENDA

1. DETERMINANTS OF OPERATING INCOME
2. BREAK-EVEN ANALYSIS
3. FIXED AND VARIABLE COSTS
4. BREAK-EVEN POINT
5. OPERATING RISKS
6. OPERATING ELASTICITY
7. PROFIT POINT
8. SUMMARY OF MAIN POINTS



Actual volumes, Costs, Revenues



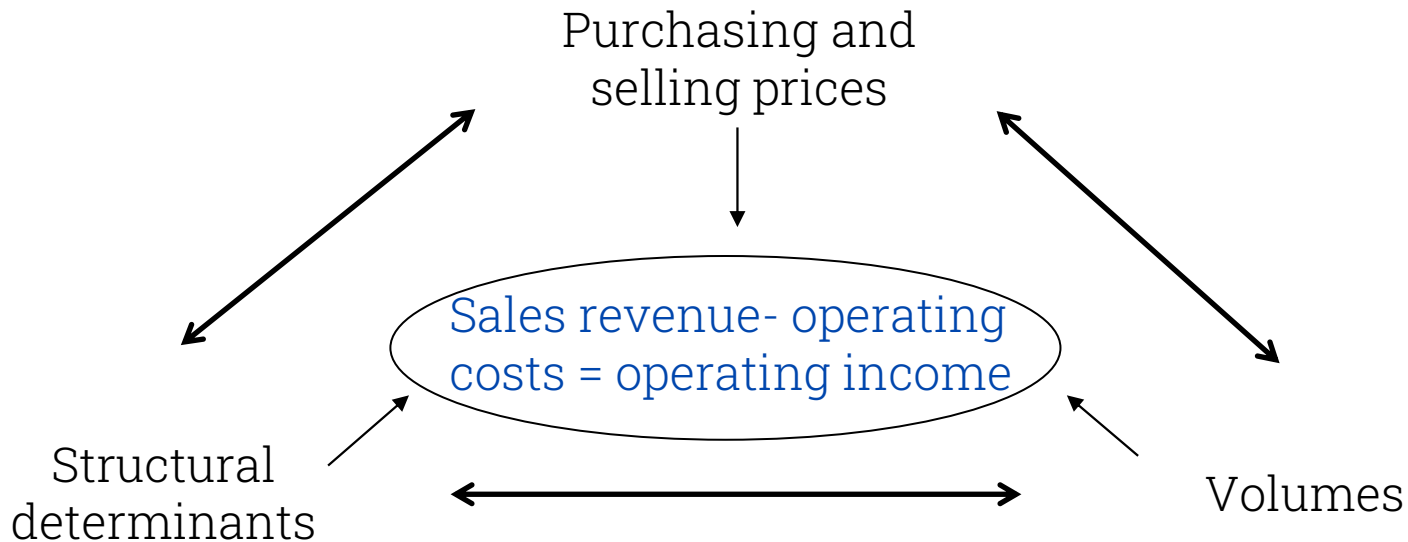
1.0

DETERMINANTS OF OPERATING INCOME



Universit 
Bocconi
MILANO

How is profitability affected by different operational choices?



Structural determinants

PRODUCTION CAPACITY

concept of maximum capacity

ECONOMIES OF SCALE

ECONOMIES OF LEARNING

ECONOMIES OF SCOPE

McDonald's produce both hamburgers and French fries at a lower average expense than what it would cost two separate firms to produce the goods separately

INDUSTRY

VERTICAL INTEGRATION

(i.e. Ikea purchasing forests in Romania to supply its own raw material) / horizontal integration (i.e. Marriott acquired Sheraton)

PRICE LEVELS

Purchasing prices: price of inputs bought from suppliers. Selling prices: price of outputs sold to consumers

VOLUMES

Different volumes influence operating profits by changing both total costs (via variable costs) and revenues



Price levels

SUPPLIERS



FOCAL COMPANY



RETAILERS



CLIENTS



Purchasing prices and Selling prices depend on internal factors (e.g. power of the firm in negotiating with suppliers or setting selling prices) and external factors (e.g. industry trends, competition levels).



2.0

BREAK-EVEN ANALYSIS



Università
Bocconi
MILANO

BREAK-EVEN ANALYSIS

BEA illustrates and models the relation between volumes produced and sold by a firm, and its operating income.



- What is the relationship between volumes sold and economic result?
- What is the effect of fixed-costs absorption?
- If sales volumes increase, how this will affect earnings?
- What is the minimum volume that needs to be sold to cover all costs?
- What effect do decisions to internalize/externalize have on cost structure, earnings, etc.?



3.0

FIXED AND VARIABLE COSTS



Università
Bocconi
MILANO

VARIABLE COSTS

Strong and direct link with production and sales value (e.g., sales commission, raw materials, external production processes).



Variable costs

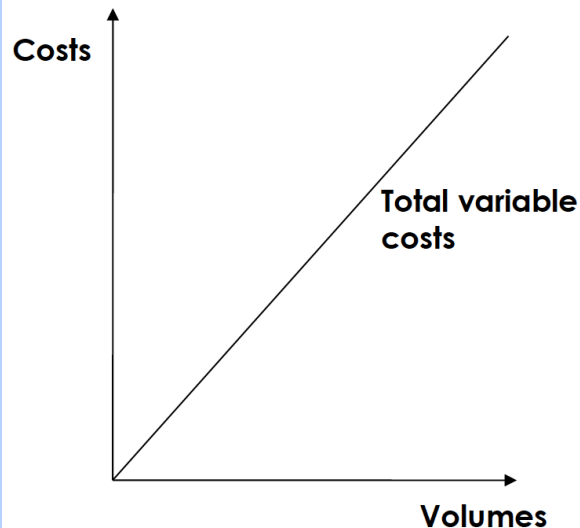
Raw material



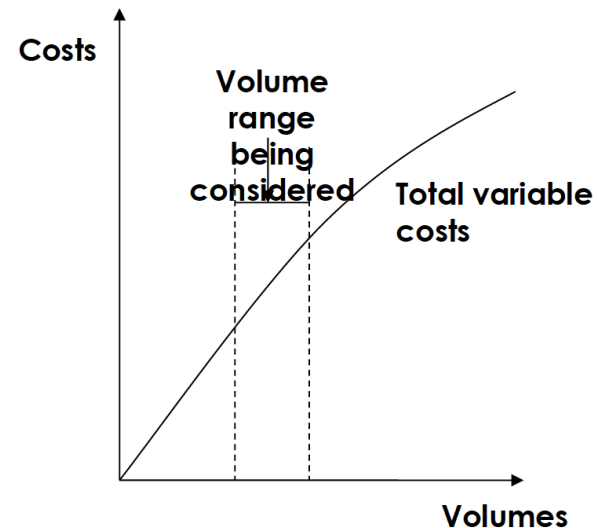
Sales commissions



Delivery charges



a. linear relationship variable costs/volumes



b. actual relationship variable costs/volumes



FIXED COSTS

NO strong and direct link with production and sales value (e.g., rent, depreciation, legal and administrative consulting)

Application: Structural fixed costs (connected to the PC); Development fixed (costs support current activities and future development – R&D, Advertising)

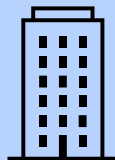


Fixed costs

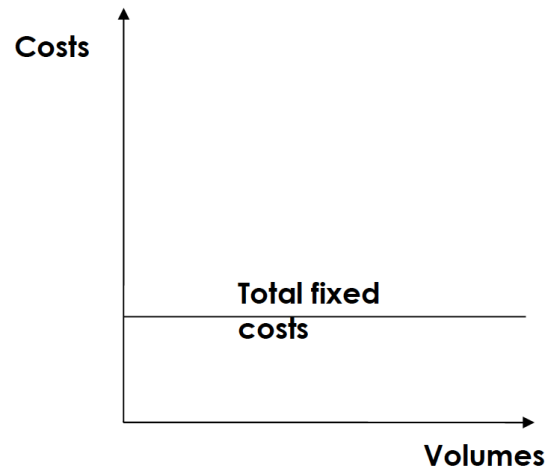
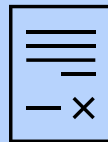
Maintenance



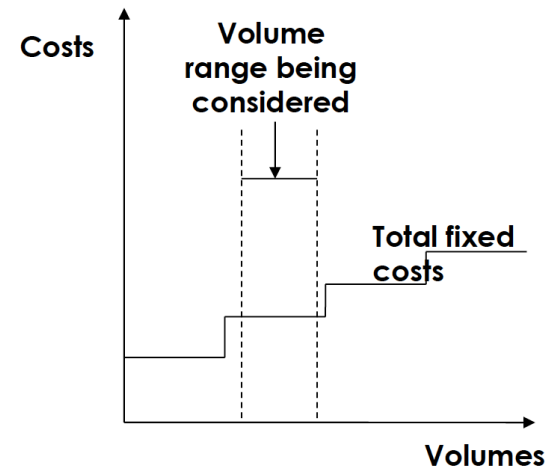
Rent



Legal consulting



a. linear relationship fixed costs/volumes



b. actual relationship fixed costs/volumes



Labour Costs

If labor can be easily increased/decreased or reallocated across business units, then labor would be a variable cost



In practice, labor is regulated and labor laws create hiring/firing costs

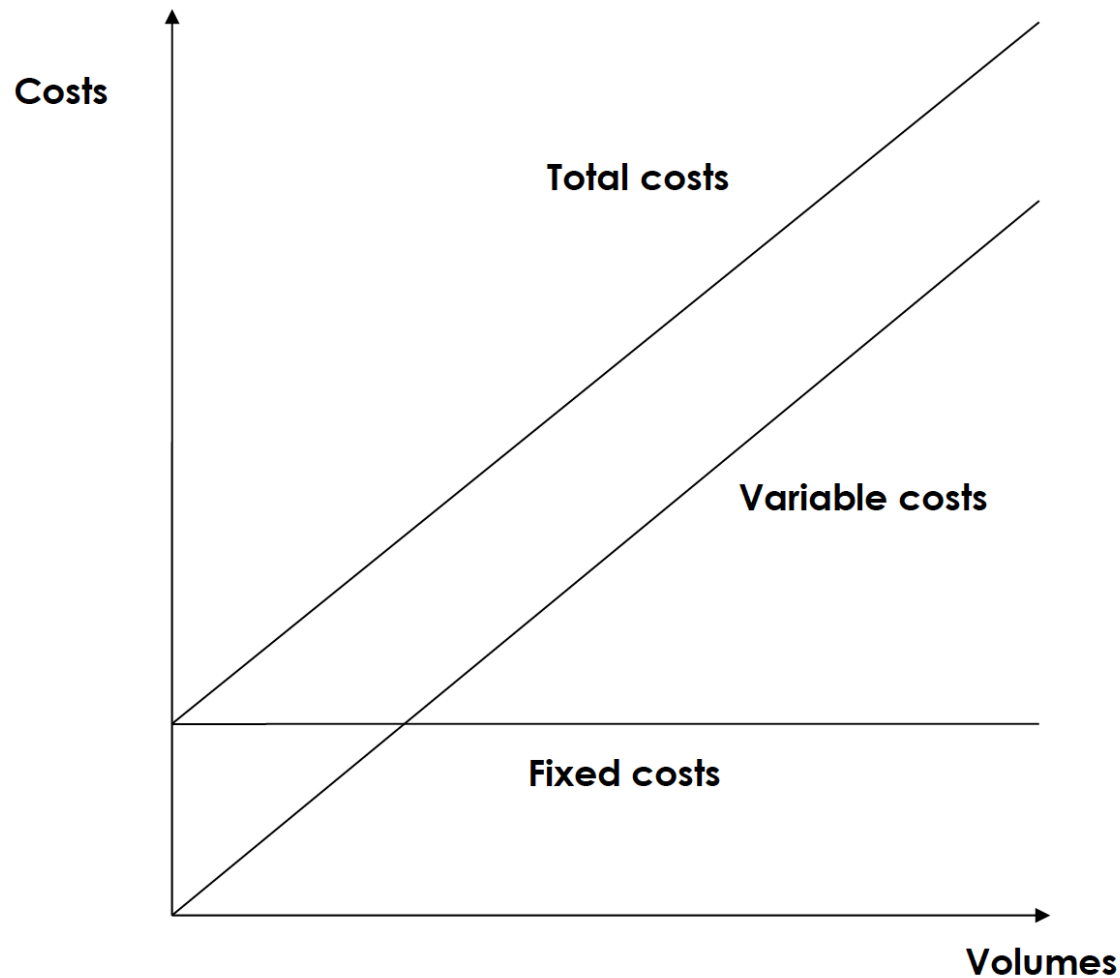


In this case, labor costs entail fixed cost components...



Total costs (TC)

$$TC = VC + FC$$



Poll time!

Maintenance cost is a:

- A) Variable cost
- B) Fixed cost



Poll time!

Direct labour cost is a:

- A) Variable cost
- B) Fixed cost



Poll time!

Administrative cost is a:

- A) Variable cost
- B) Fixed cost



Poll time!

Cost of material is a:

- A) Variable cost
- B) Fixed cost



4.0

BREAK-EVEN POINT



Università
Bocconi
MILANO

OPERATIVE BREAK-EVEN POINT

Amount of sales that allows the firm to cover its operating costs.



Operative BEP in Quantity (QBEP)

Operative BEP, in quantity (QBEP), is the production volume for which total revenues (R) are equal to total costs (TC)

$$R = TC$$

$$R = VC + FC$$

$$R_u \times Q_{BEP} = VC_u \times Q_{BEP} + FC$$

$$Q_{BEP} (R_u - VC_u) = FC$$

$$Q_{BEP} = FC / (R_u - VC_u)$$

$$CM_u = R_u - VC_u$$

$$Q = FC / CM_u \text{ Unit Contribution Margin}$$

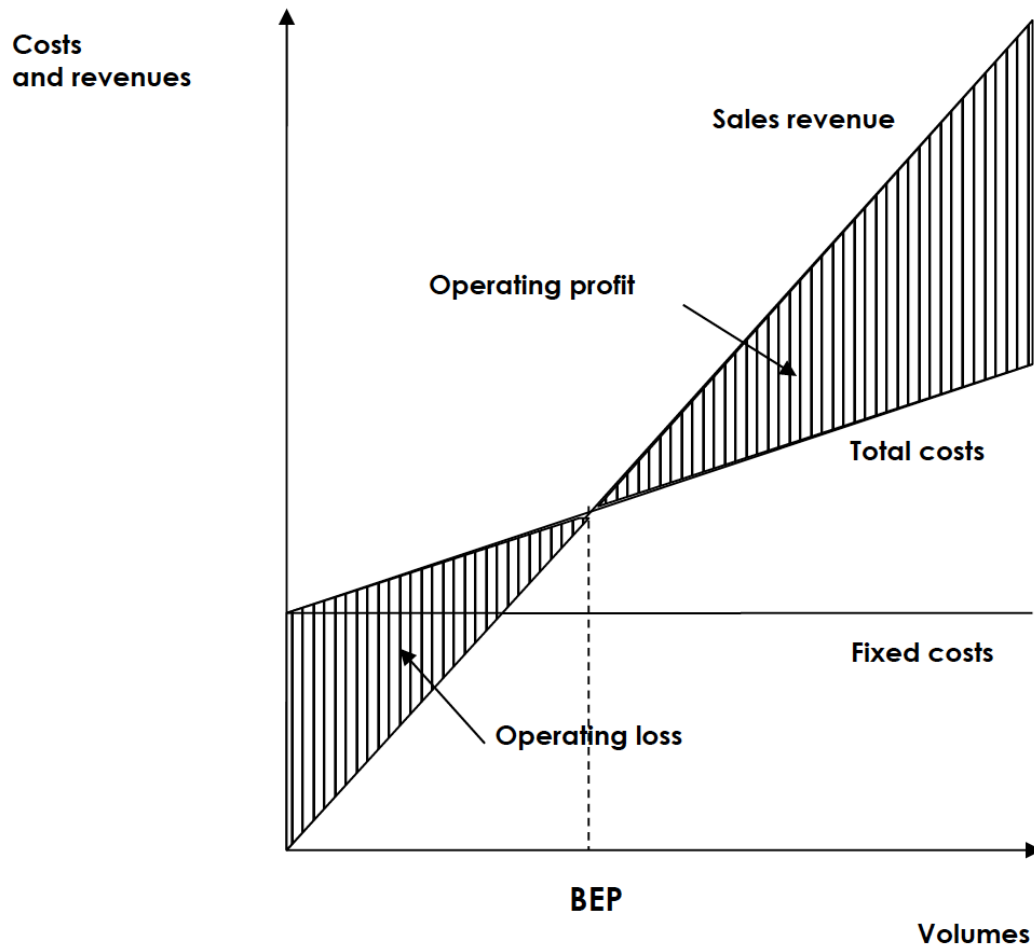


Break-even point

At BEP firms start to make an operating profit



Università
Bocconi
MILANO



Break-Even Point

CONTRIBUTION MARGIN

as a % of revenue (CM%, or CM ratio)

$$CM\% = C_{mu} / R_u$$

$$CM\% = CM / R$$

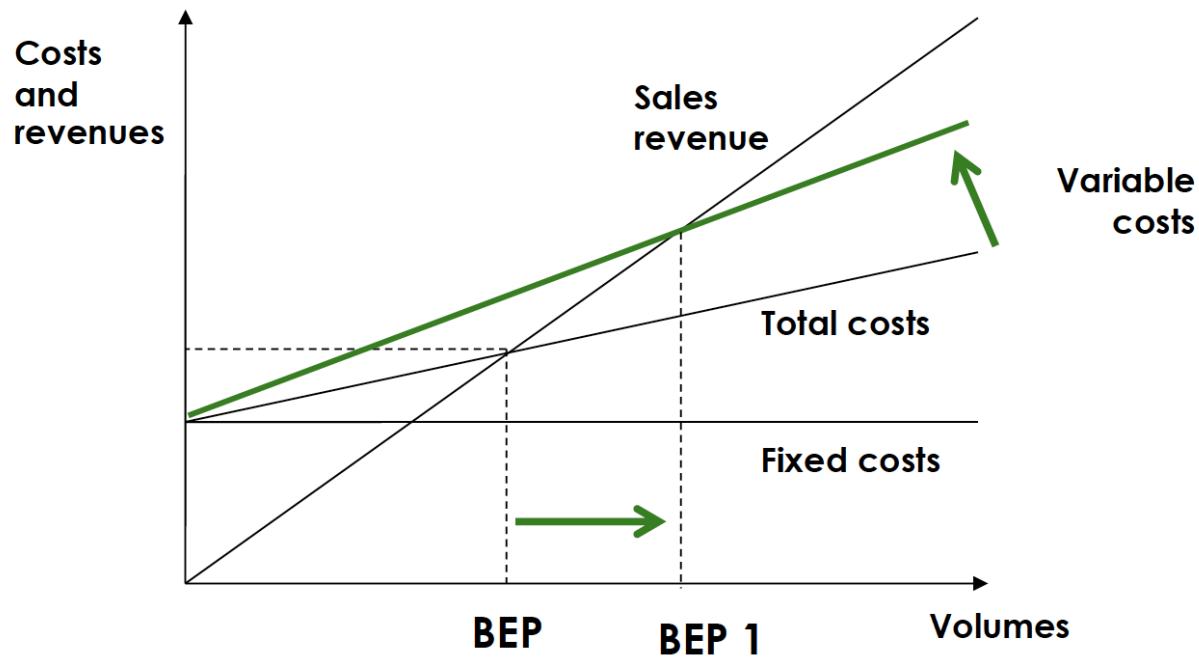
BREAK-EVEN POINT IN REVENUES (SALES VERSION)

Calculated with the percentage contribution margin

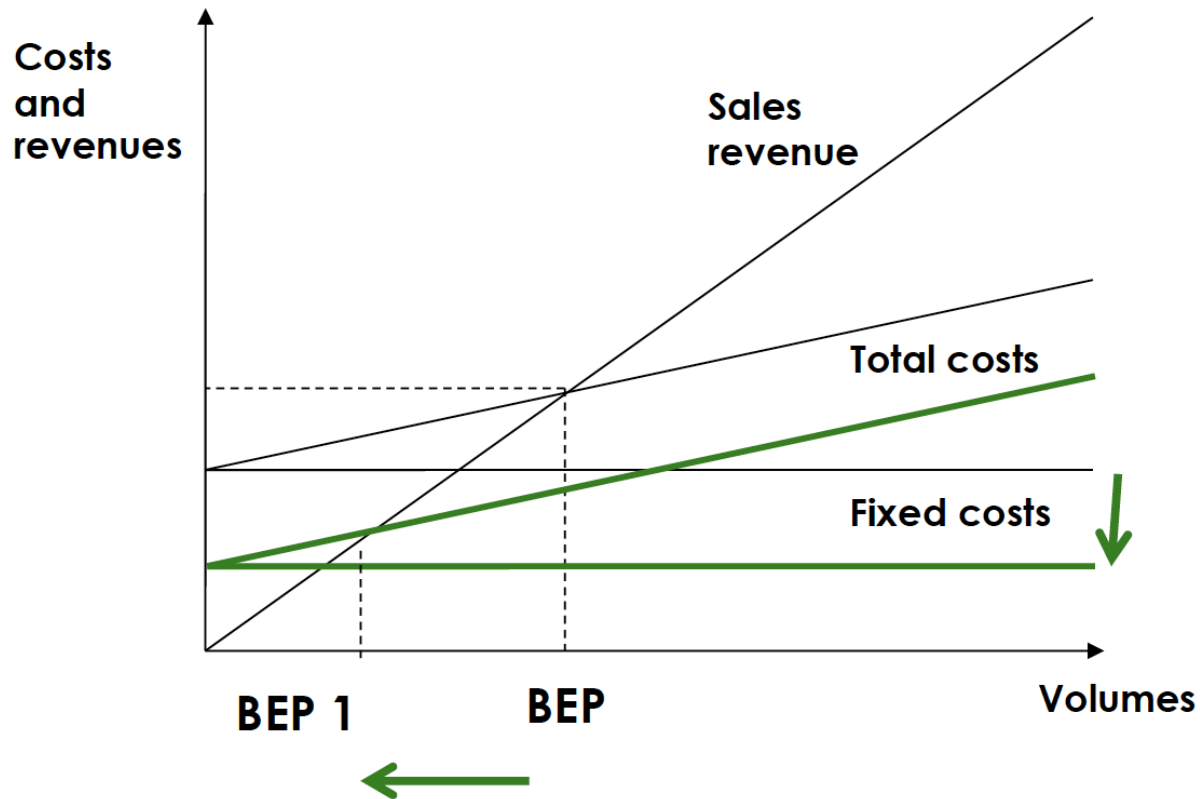
$$R \text{ BEP (in Revenues)} = FC / CM\%$$



Break-even point if ... Variable Costs increase



Break-even point if ... Fixed Costs decrease



Let's organize a party – how many tickets to BEP?

FIXED COSTS

Hall (400p)	2000€
Dee-jay	300€
Lights + Decoration	200€
Security	400€
Cleaning	300€
Audio System	500€
Barman	300€

VARIABLE COSTS

Beer	3€ x person
Sparkling Wine	4€ x person
Spirits	2,5€ x person
Soft Drinks	2,5€ x person
<u>Panettone</u>	2€ x person

Ticket price per person = 30 €



Let's organize a party – how many tickets to BEP?

FIXED COSTS

Hall (400p)	2000€
Dee-jay	300€
Lights + Decoration	200€
Security	400€
Cleaning	300€
Audio System	500€
Barman	300€

VARIABLE COSTS

Beer	3€ x person
Sparkling Wine	4€ x person
Spirits	2,5€ x person
Soft Drinks	2,5€ x person
<u>Panettone</u>	2€ x person

Ticket price per person = 30 €



5.0

OPERATING RISKS



Università
Bocconi
MILANO

OPERATING RISKS

Operating risks is linked to two components of the organization's economic structure: BEP and degree of the operating leverage.

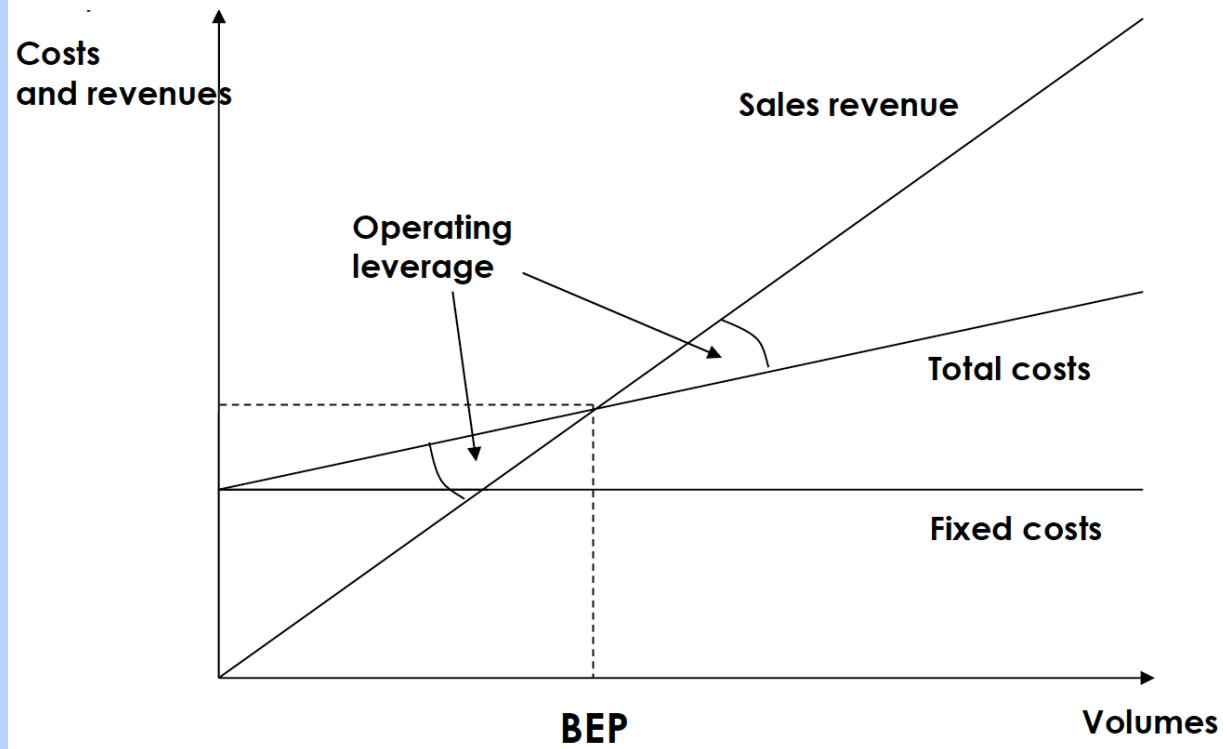


OPERATING LEVERAGE

Operating leverage is the size of the differential (gap) between revenues and total costs above and below the break-even point.



Operating risks



Operating risks

Suppose that there are exogenous changes in volume:

FOR A FIRM WITH A VERY RIGID COST STRUCTURE

i.e. high ratio of fixed costs to total costs

- The firm will react badly to drops in volumes (less room to distribute fixed costs over units of output)
- The firm will respond positively to increases in volumes

FOR A FIRM WITH A FLEXIBLE COST STRUCTURE

i.e. low ratio of fixed costs to total costs

- If volumes drop, the firm can easily reduce costs
- If volumes increase, the firm experience unavoidable cost increases



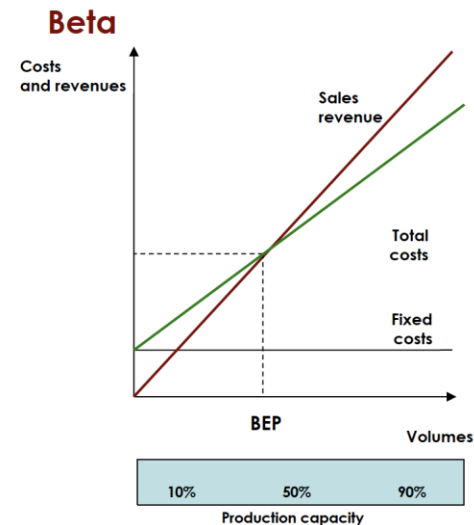
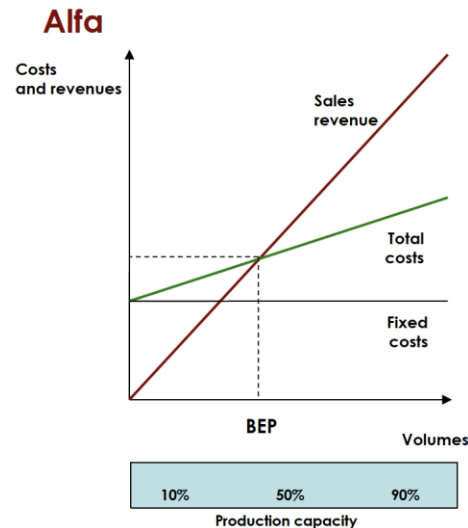
The Elements of Operating risks

THE LEVEL OF BEP

(the higher the BEP, the higher the risk)

THE OPERATING ELASTICITY

the relationship between total variable costs and fixed costs at the BEP (the higher the elasticity, the lower the risk)



Operating risks

1. Operating risk is not necessarily a bad thing: it amplifies losses (as we move in the area left to the BEP), but it amplifies profits (as we move in the area right to the BEP).
2. The choice of, e.g. two plants with same BEP but different operating risk depends on our estimate of how much volumes sold would exceed the BEP, and on the manager's degree of risk aversion.



6.0

OPERATING ELASTICITY



Università
Bocconi
MILANO

OPERATING ELASTICITY

Operating elasticity can be measured by calculating the relationship between total variable costs and fixed costs at the BEP. The higher the operating elasticity, the lower the risk.



Operating elasticity: an application

Graphically, it reflects into the gap between the lines of total revenues and costs. The wider the wedge between the lines, the higher the risk.

	PLANT A	PLANT B
Price	12	12
VCu	1.30	1.20
FC	10,000	15,000
BEP		
Operating Elasticity		



Which option do you prefer between A and B?



Operating elasticity: an application

	PLANT A	PLANT B
Price	12	12
VCu	1.30	1.20
FC	10,000	15,000
BEP	$10,000 / (12 - 1.30)$ $= 934,57$	$15,000 / (12 - 1.20)$ $= 1,388.88$
Operating Elasticity	$(1.30 * 934,57)$ $/ 10,000 = 0.121$	$(1.20 * 1,388.88)$ $/ 15,000 = 0.111$



7.0
PROFIT POINT



Università
Bocconi
MILANO

Profit Point

1. Profit point = sales volume that covers all costs (operations + financial charges and fiscal costs) and provides an acceptable net income.
2. Add the target “operating income” to fixed costs.

$$\text{Sales} = \frac{\text{FC} + \text{Target Operating Income (TOI)}}{\text{CM\%}}$$

3. The “income” that covers costs of non-core operations and builds up a desired net profit.



Profit Point: example 1

$TFC = 10,000\text{€}$

$R_u = 5\text{€}$

$VC_u = 2\text{€}$

What is the output necessary to earn 5,000 € as target net income.

Volumes= _____



Profit Point: example 1

$$\text{TFC} = 10,000\text{€}$$

$$\text{Ru} = 5\text{ €}$$

$$\text{VCu} = 2\text{ €}$$

What is the output necessary to earn 5,000 € as target net income.

$$\text{Volumes} = \frac{10,000\text{ €} + 5,000\text{ €}}{3} = 15,000$$



Profit Point: example 2, food store (first year activity)

- Average prize for a meal 16,00 €. (Ru)
- Average cost for a meal 4,00 € (VCu)
- Total fixed costs: 405,000 € (FC)
- Financial charges: Loan of 200,000€ at 12% of interest rate
- Taxes unrelated to income: 6,000 €
- Income taxes: 50% of income before taxes
- Target net income 75,000 €



Profit Point: example 2, food store (first year activity)

Revenues = _____ =

Revenues = _____ =

Volumes = _____ =

Volumes = _____ =



Profit Point: example 2, food store (first year activity)

- Average prize for a meal 16,00 €. (Ru)
- Average cost for a meal 4,00 € (VCu)
- Total fixed costs: 405,000 € (FC)
- Taxes unrelated to income: 6,000 €
- Financial charges: Loan of 200,000€ at 12% of interest rate
- Income taxes: 50% of income before taxes
- Target net income 75,000 €

Financial charges (24,000€) + Non-income taxes (6,000€)
+ Income before taxes (150,000€) = TOI 180,000€



Profit Point: example 2, food store (first year activity)

$$\text{Revenues} = \frac{\text{FC} + \text{TOI}}{\text{CM}\%} =$$

$$\text{Revenues} = \frac{405,000 \text{ €} + 180,000 \text{ €}}{0,75 \text{ } (= (\text{Ru} - \text{VCu}) / \text{Ru})} = 780,000 \text{ €}$$

$$\text{Volumes} = \frac{\text{FC} + \text{TOI}}{\text{CMu}} =$$

$$\text{Volumes} = \frac{585,000 \text{ €}}{12} = 48,750 \text{ meals}$$



8.0

SUMMARY OF MAIN POINTS



Università
Bocconi
MILANO

In summary

- Understand the structural determinants of the operating income.
- Fixed vs Variable costs.
- The link between operating risks and BEP
- Economic meaning and computation of Operating elasticity
- Computing the profit point.

THANK YOU



Università
Bocconi
MILANO